HappyMaxx Consulting

World Happiness EDA Report

2015 – 2019 · 156 Countries · 782 Observations

Data-driven decisions for happier societies

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Who Are We: HappyMaxx Consulting

Our Mission

HappyMaxx Consulting helps governments understand what makes people happier and which policies improve quality of life. Using global happiness data, we identify key factors like healthcare, economic stability, and social support, linked to higher wellbeing. Our goal: help countries make data-driven decisions that create happier and healthier societies.

Diagnose

Identify the drivers of low happiness in your country using 5 years of validated global data

Prioritise

Rank policy levers by their proven impact on wellbeing so budgets go where they matter

Track

Measure year-on-year progress with the same methodology used by the UN World Happiness Report

Our Three Research Hypotheses

H1

Economic & Social Development

Countries with higher GDP per capita, stronger social support, and higher life expectancy will report higher happiness scores.

H2

Demographics & Happiness

Population size, density, and median age may influence happiness; More stable, developed populations expected to score higher.

H3

Profile of High-Happiness Countries

Top-scoring countries share common traits: strong economies, low corruption, high freedom, and trusted institutions.

Data Sources, Database & Cleaning

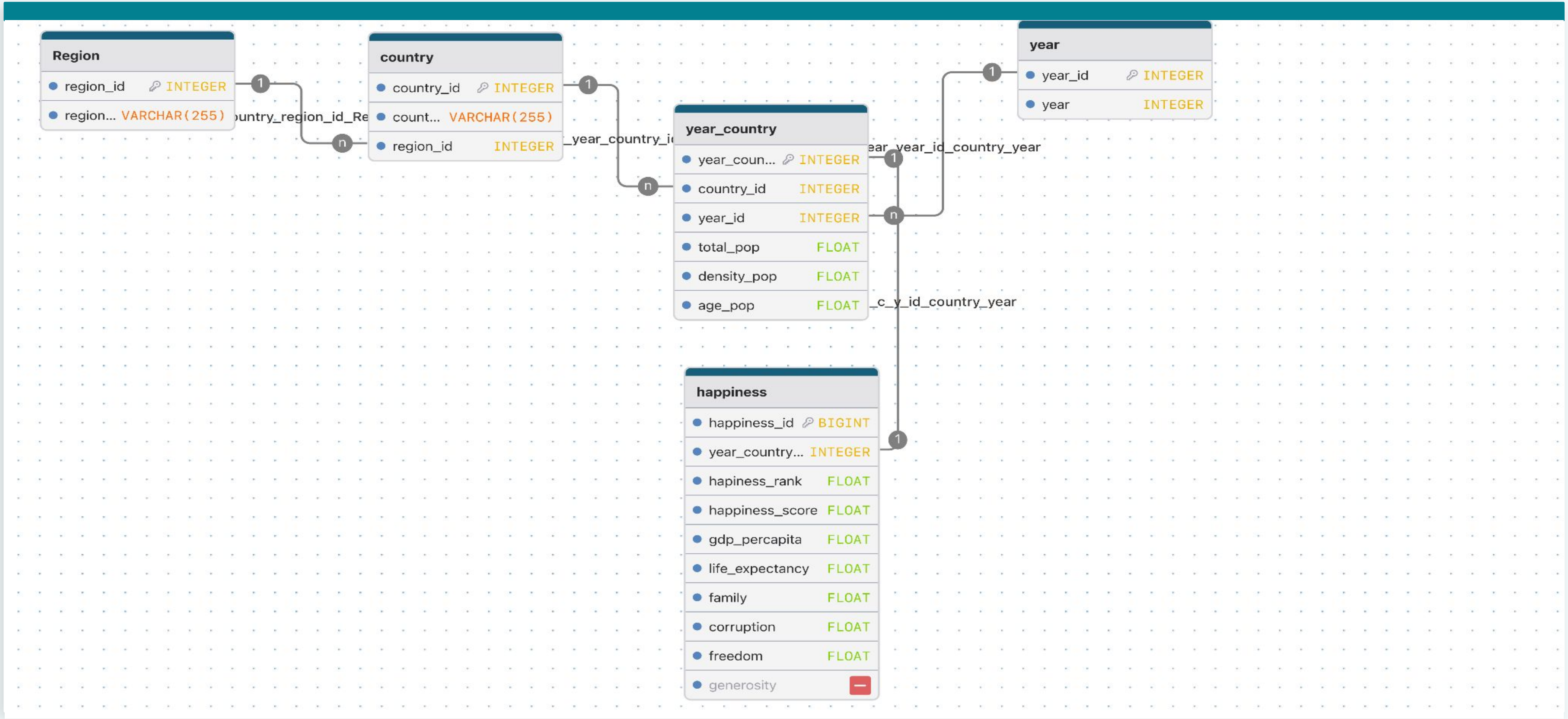
Sources & Schema

- 📄 World Happiness Report (WHR) 2015–2019
- 🌍 UN World Population Prospects (population data)
- 🗄️ MySQL schema: region → country → year_country → happiness
- 📐 5 normalised tables with FK constraints & CASCADE rules
- 📊 782 rows · 14 features · 0 nulls after cleaning

Cleaning Steps

- 🔗 Merged happiness + population on country key & year
- ✏️ Manual aliases: North Cyprus→Cyprus, Somaliland→Somalia, Macedonia→North Macedonia
- 🧮 Country-year aggregation via groupby (mean / mode)
- 🩺 1 null: UAE 2018 corruption filled with country median (outlier-robust)
- ✅ Final dtypes verified: int64, float64, str — all correct

Creating Our Database - ERD



Dataset at a Glance — Key Statistics

5.41	5.37	1.11	7.63	2.69
Global mean happiness score	Median happiness score	Std deviation (score spread)	Highest score (Finland)	Lowest score (South Sudan)

Pearson r with Happiness Score

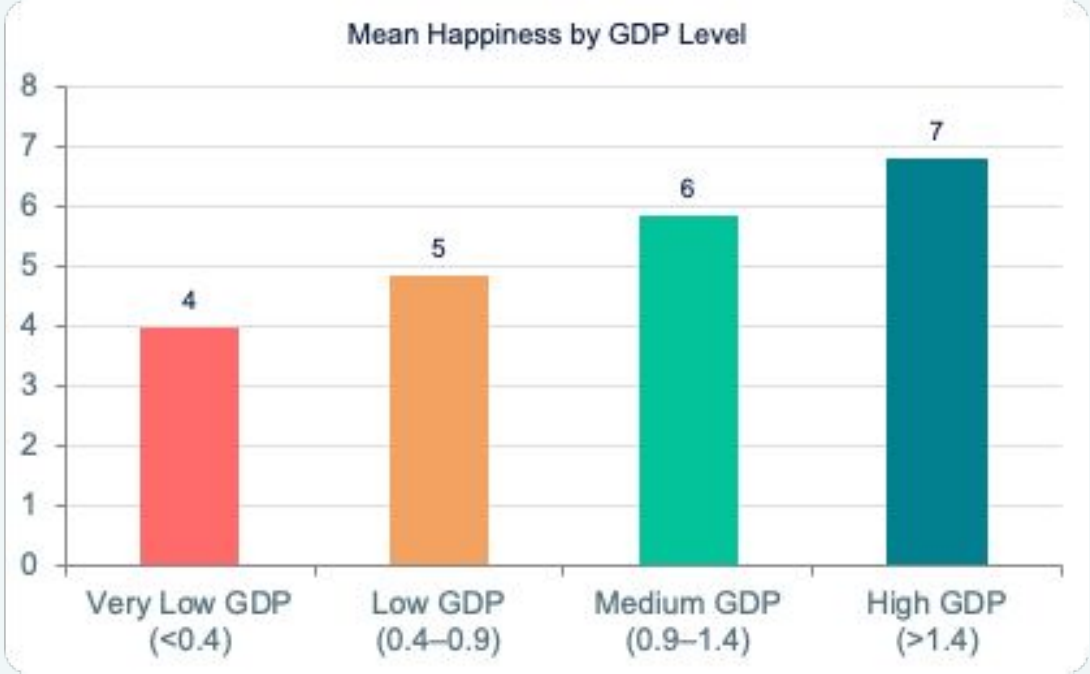
Variable	r	Strength
GDP per capita	**0.79**	Strong ✅
Life expectancy	0.74	Strong ✅
Median age	0.68	Moderate–Strong ✅
Family / social support	0.65	Moderate ✅
Freedom	0.55	Moderate ✅
Gov. trust (low corruption)	0.40	Moderate ✅
Generosity	0.14	Weak ⚠️
Pop. density	0.08	Negligible ❌
Total population	–0.04	Negligible ❌

Key takeaway

Economic & social factors dominate. Population size is irrelevant.

H1 · GDP per Capita vs Happiness Score

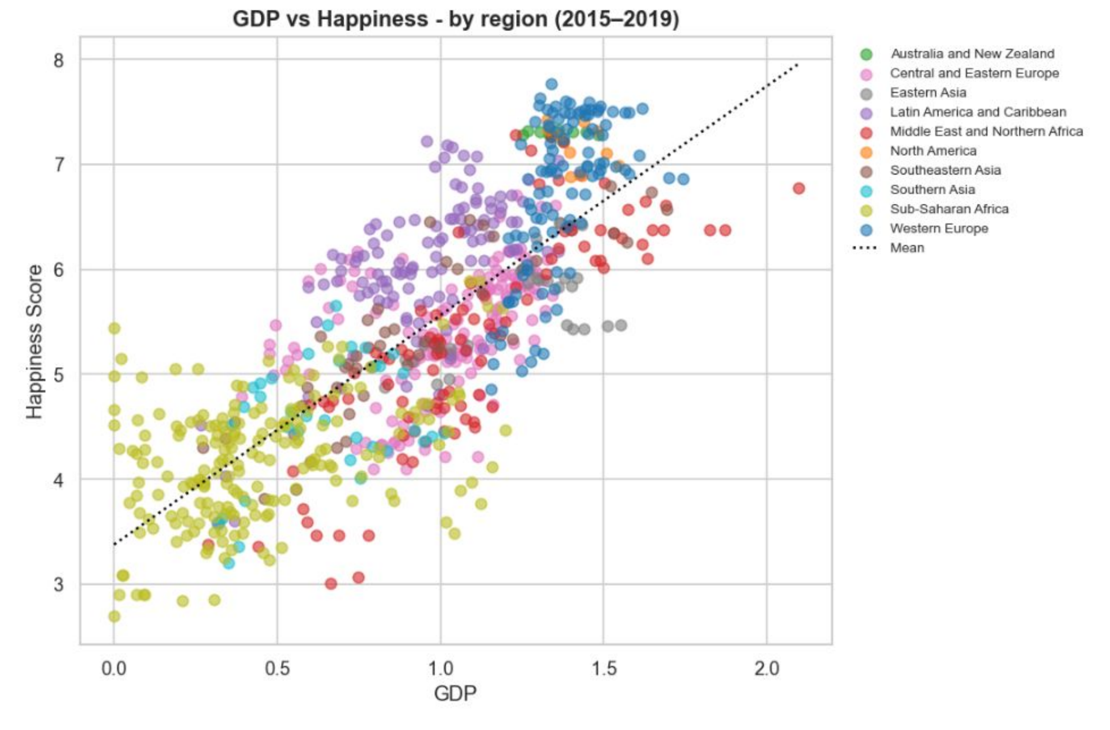
r = 0.79 — Strongest predictor in the dataset



Findings

- High-GDP countries score 6.8 vs 4.0 for very-low-GDP
- 1.7× happiness gap between richest and poorest GDP tier
- Western Europe & North America dominate the top rankings
- Richer countries also show higher life expectancy (r=0.77 between GDP & life exp.)

Outliers

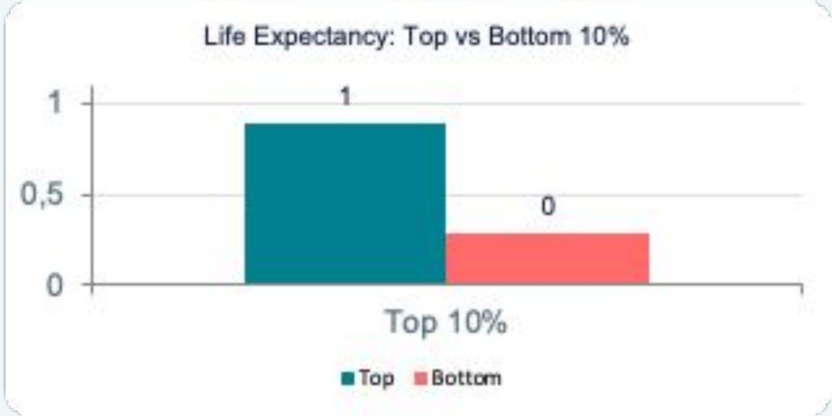


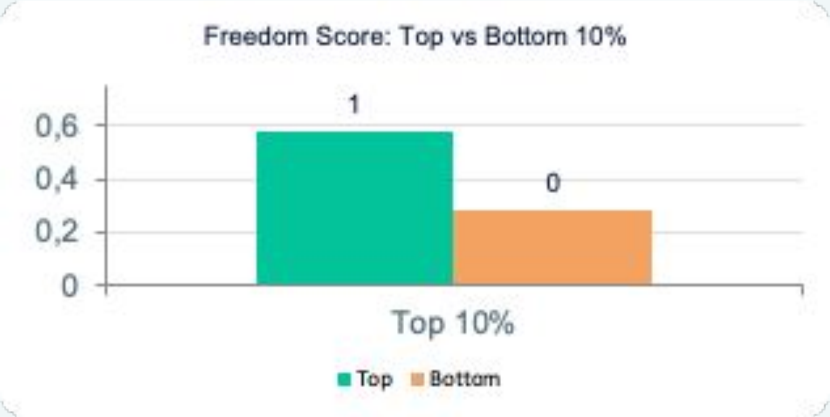
Findings

- Sub-Saharan countries and Latin America are a positive outlier — scores above its GDP level due to strong social ties

- Middle-east countries, eventhough they are rich, are no as happy as they are supposed to be

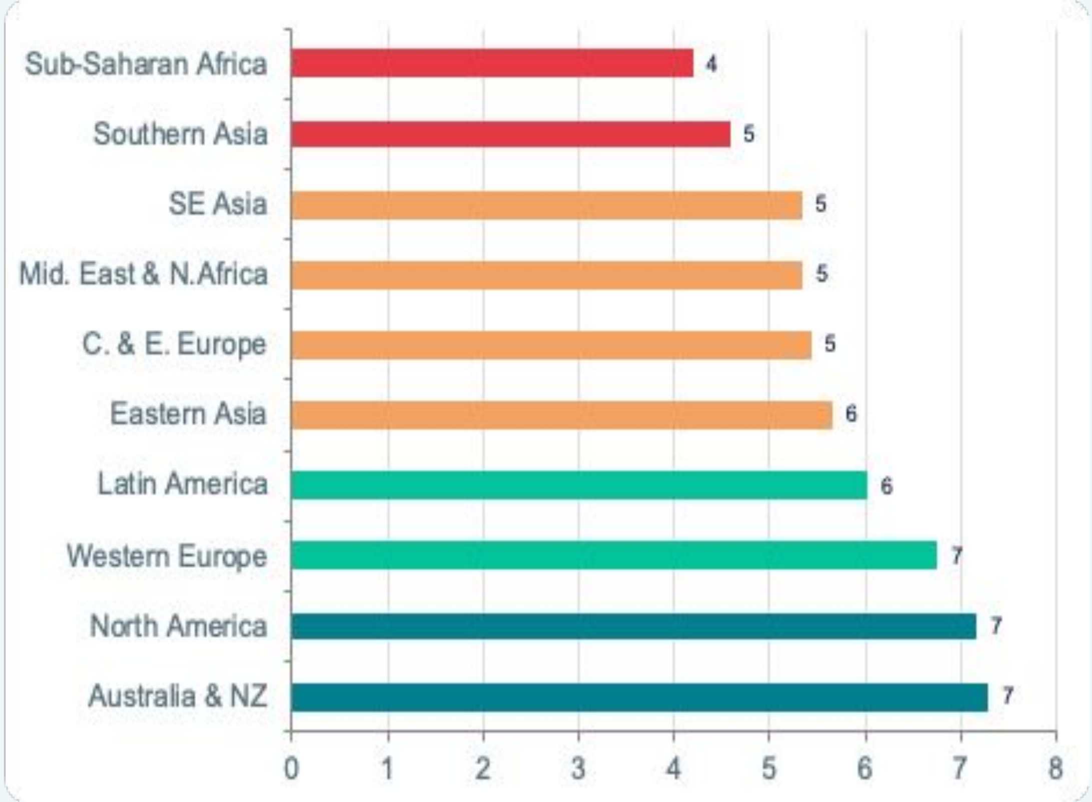
H1 · Life Expectancy & Freedom Also Drive Happiness





Variable	r with Happiness	Top 10% mean	Bottom 10% mean
Life expectancy	0.74	0.885	0.292
Freedom	0.55	0.582	0.284
Family / social support	0.65	1.384	0.648

Happiness Across the World — Regional Rankings



Top Region

Australia & NZ avg 7.30

Bottom Region

Sub-Saharan Africa avg 4.19

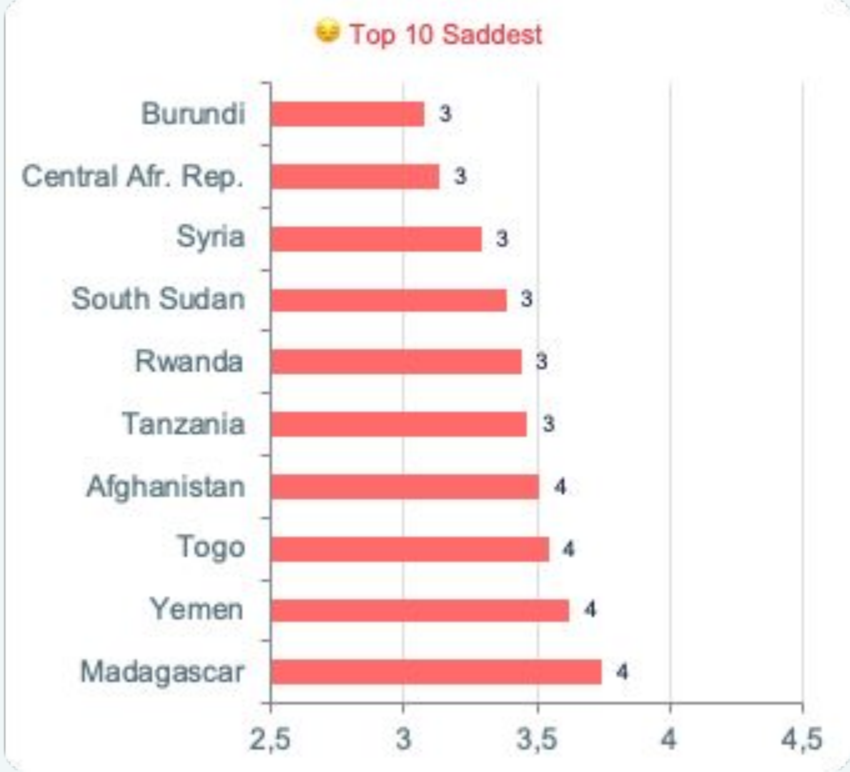
Global Mean

5.41 across all regions

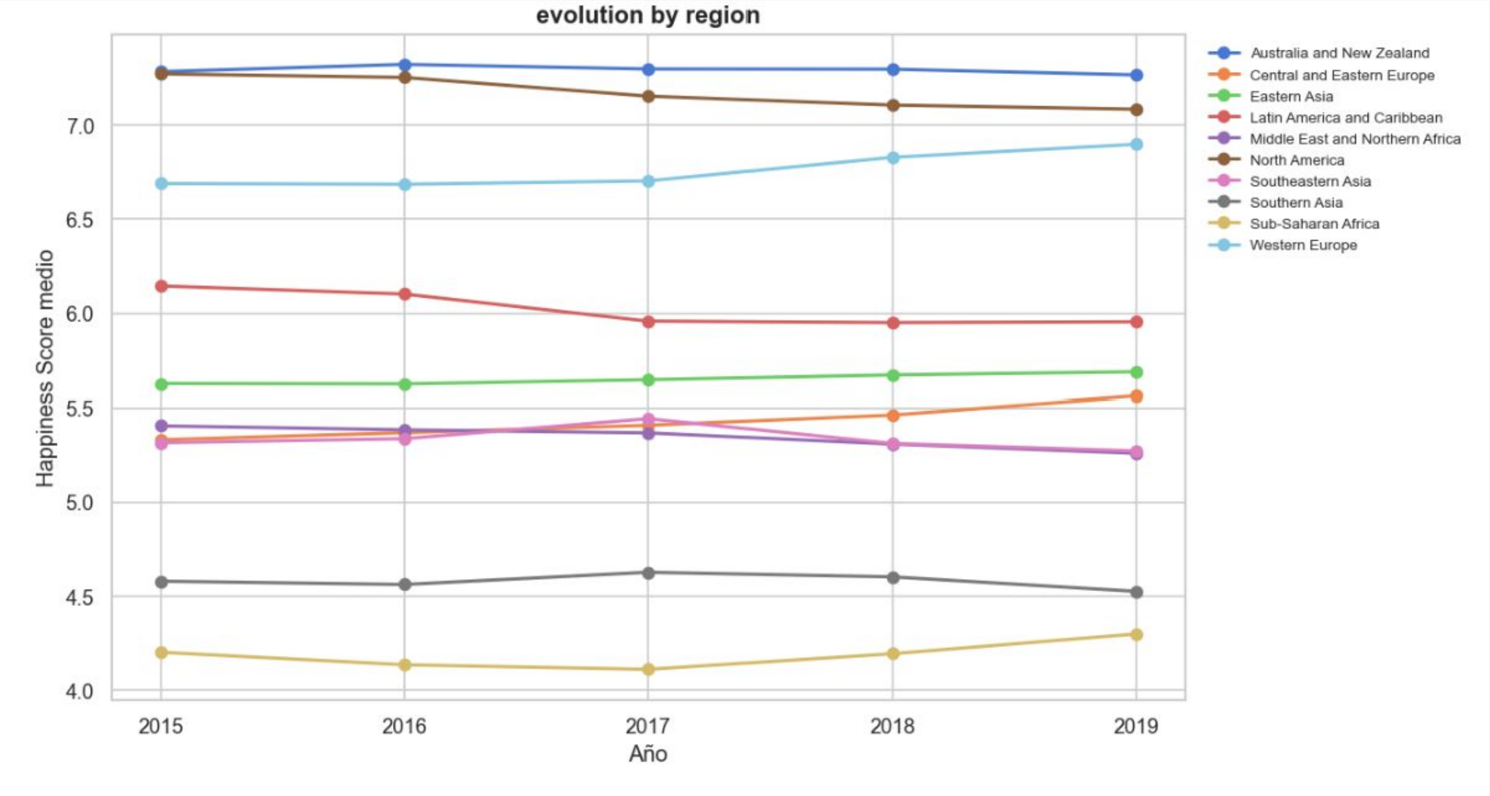
Biggest Gap

3.1 points top vs bottom

Top 10 Happiest & Saddest Countries (2015-2019)

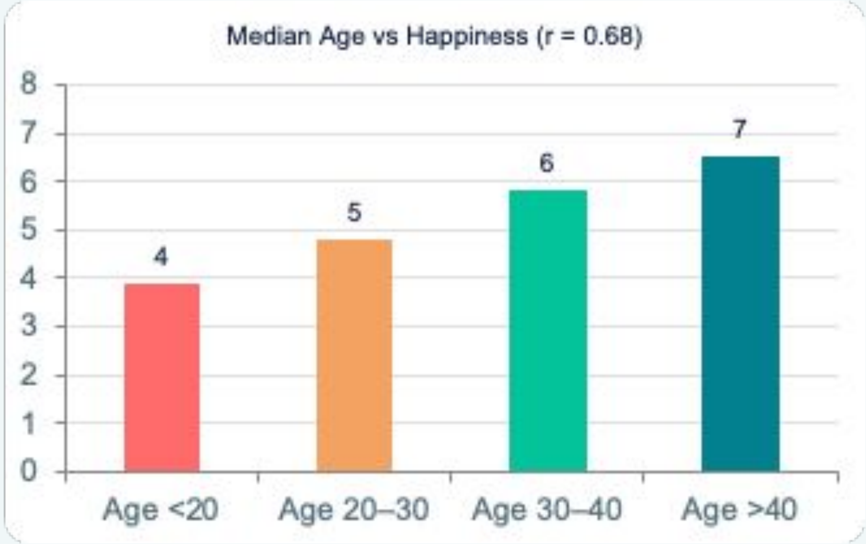


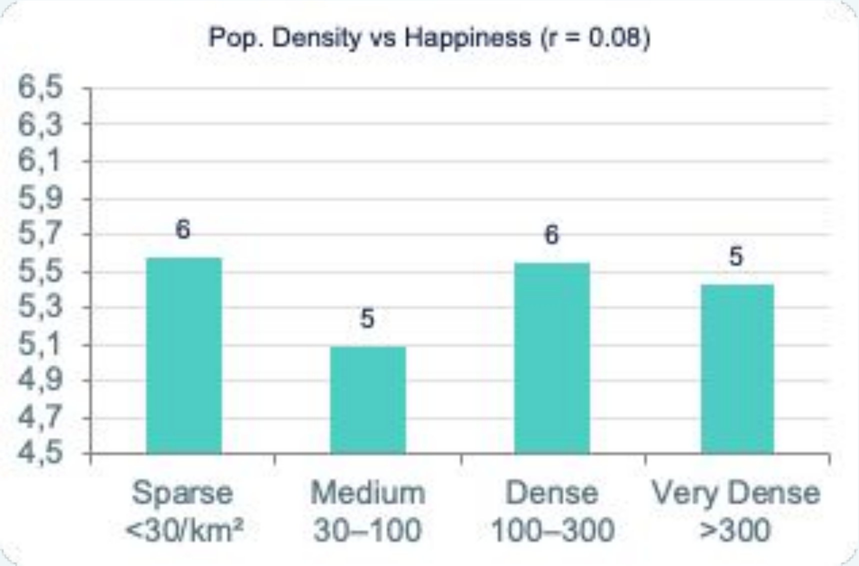
Happiness Over Time — Is the World Getting Happier?



💡 Global mean happiness remained flat (≈5.37–5.41) over 2015–2019. No region showed dramatic improvement. Western Europe trended slightly upward; North America dipped after 2016.

H2 · Demographics & Happiness





Median age (r = 0.68)

MODERATE LINK ✅

Older, demographically mature populations score significantly higher. Age is a strong proxy for development level.

Total population (r = -0.04)

NO LINK ❌

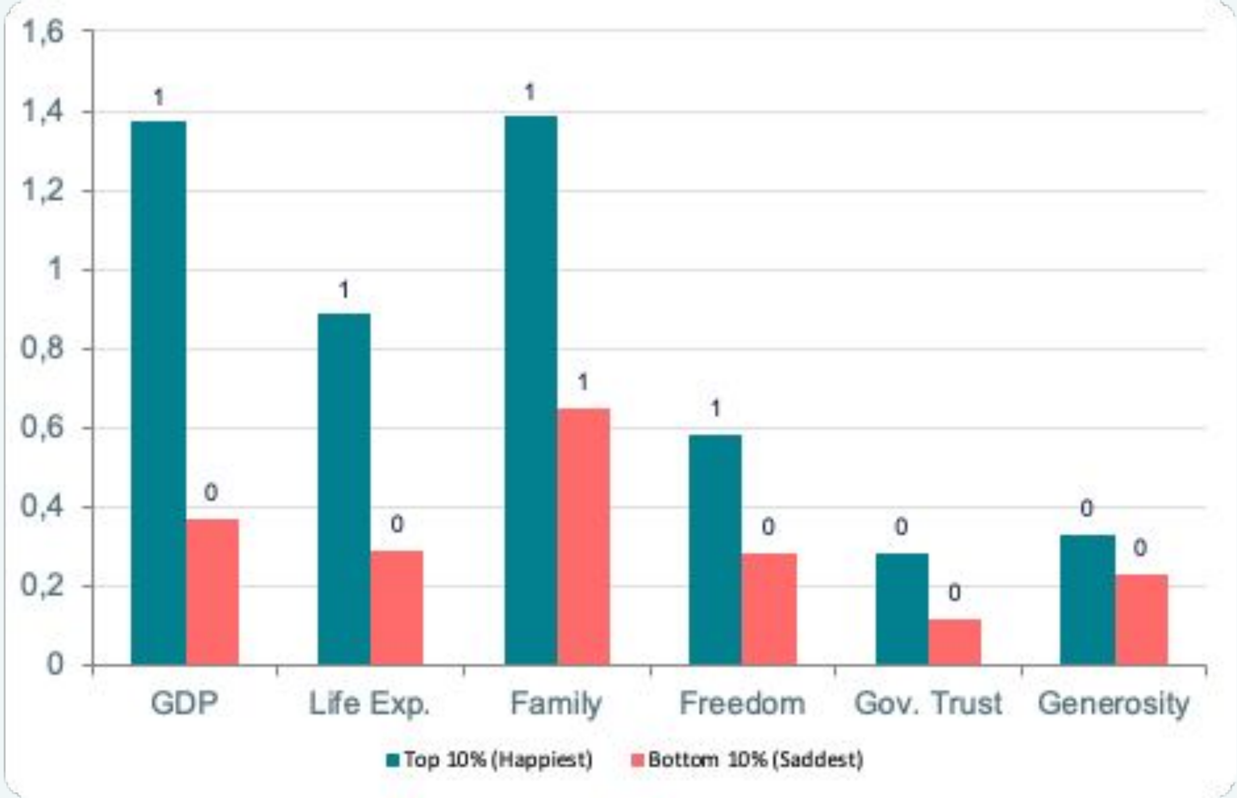
Population size has virtually zero relationship with happiness. Big countries are not happier countries.

Pop. density (r = 0.08)

NO LINK ❌

Density is irrelevant. Singapore & Netherlands rank highly despite extreme density.

H3 · The 'Dream Country' Profile — Top 10% vs Bottom 10%



Top ÷ Bottom

GDP	3.7×
Life Exp.	3.0×
Family	2.1×
Freedom	2.0×
Gov. Trust	2.4×
Generosity	1.4×

SQL Insight — Do Richer Countries Live Longer?

Obviously yes and the gap is striking!

3.0×

Higher life exp.
in richest vs poorest GDP tier

0.835

Avg life exp. score
high-GDP countries

0.282

Avg life exp. score
very-low-GDP countries

Conclusions

H1 — CONFIRMED ✅

Economic & social development drives happiness

GDP (r=0.79), life expectancy (r=0.74), and family support (r=0.65) are the strongest predictors. High-GDP countries score 6.8 vs 4.0 for the lowest tier. Latin America is a notable outlier, scoring above its GDP level.

H2 — PARTIALLY REJECTED ⚠️

Population size & density are irrelevant; median age matters

Total population (r=–0.04) and density (r=0.08) show negligible correlation. However, median age (r=0.68) is a meaningful predictor — it reflects decades of investment in stability and health.

H3 — CONFIRMED ✅

Happy countries share a clear structural profile

Top 10% vs bottom 10%: GDP is 3.7× higher, gov. trust 2.4× higher, life expectancy 3× higher. The recipe: wealthy + healthy + free + trustworthy institutions. No exceptions among the top 10.

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Thank You

Maximising Happiness, One Data Point at a Time

Recommended Next Steps

- 🔬 Run regression models to quantify each variable's independent contribution controlling for GDP
- 🗺️ Add geospatial layer — map happiness to visualise regional clusters interactively
- 📅 Extend dataset post-2019 to capture COVID-19 impact on global wellbeing
- 🏛️ Build policy simulator: model which interventions move a country up the happiness ladder